

Project 1A: Data Analyst Securitisation

Professional Power BI Dashboard Documentation

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Project Domain: BFSI / Structured Finance / Credit Risk Analytics

Tool Used: Microsoft Power BI, Power Query and DAX

Submission Type: Private Project Work Documentation

This document presents a comprehensive explanation of the securitisation risk assessment dashboard developed for Project 1A. The report describes the business problem, data sources, dashboard pages, analytical methodology, DAX logic, key insights, and recommendations. The dashboard has been designed to simulate a real-world bank or NBFC credit-risk reporting environment covering portfolio monitoring, IFRS 9 ECL analytics, stress testing, delinquency analysis, geographic concentration risk, vintage analysis, and dynamic loss performance.

Item	Description
Project Title	Data Analyst Securitisation Risk Assessment Dashboard
Primary Output	Interactive Power BI dashboard with seven analytical pages
Business Focus	Credit risk, securitisation performance, IFRS 9 ECL and stress testing
Confidentiality	Private project work; not for public code or dashboard sharing

1. Executive Summary

The securitisation dashboard was developed to provide a structured risk assessment framework for an assumed auto-loan portfolio. It combines portfolio-level KPIs, delinquency metrics, IFRS 9 ECL analysis, stress testing, geographic risk ranking, vintage pool performance, and monthly dynamic loss performance in one Power BI report.

The dashboard is designed for business users such as risk analysts, credit managers, investors, portfolio monitoring teams, and management stakeholders. It gives a quick view of portfolio health while also allowing drill-down into risk segments such as 90+ DPD loans, Stage 3 exposure, high-PD states, and stressed ECL impact.

The final report contains seven pages: Portfolio Overview, DPD and Delinquency Analysis, IFRS 9 ECL Analysis, Stress Testing and Scenario Analysis, Geographic Risk Analysis, Vintage and Static Pool Analysis, and Dynamic Loss and Performance Analysis.

Key details:

Item	Description
Dashboard Pages	7 completed analytical pages
Total EAD	Approximately 326.16M
Total ECL	Approximately 11.66M
Main Risk Driver	Stage 3 / high-PD loan segments

2. Business Problem and Project Objectives

In securitisation, investors and originators need to understand whether the underlying loan pool is performing as expected. A securitisation portfolio can appear healthy at aggregate level while hidden risk may be concentrated in certain states, DPD buckets, vintages, or IFRS 9 stages.

The main objective of this project is to build a Power BI solution that converts assumed banking or SAP/ERP-style loan data into meaningful risk intelligence. The dashboard should answer questions such as: What is the total exposure? Which regions have the highest expected credit losses? How much loss may increase under stress? Which vintages show deterioration? How are monthly defaults and recoveries behaving?

The project also demonstrates the practical importance of domain knowledge in BFSI analytics. It goes beyond basic charts by combining credit-risk concepts such as PD, LGD, EAD, ECL, delinquency buckets, pool factor, vintage seasoning, recoveries and stress scenarios.

Key details:

Item	Description
Objective 1	Build portfolio risk monitoring dashboard
Objective 2	Develop IFRS 9 ECL analytical view
Objective 3	Create stress testing and performance pages
Objective 4	Present clear insights for decision-making

3. Data Sources and Dataset Description

The solution uses four primary datasets: Auto Loan Securitisation Data, DPD Snapshot History, Static Pool Vintage Data, and Dynamic Loss Monthly Data. Each dataset contributes a different view of portfolio behavior.

Auto Loan Securitisation Data contains borrower-level and loan-level fields such as LoanID, BorrowerID, CurrentBalance, EAD, ECL_Provision, PD_Estimate, LGD_Estimate, IFRS9_Stage, DelinquencyDays, Region and State. This is the main table used for portfolio, ECL, geographic and delinquency pages.

Static Pool Vintage Data supports vintage and pool performance analysis using fields such as MonthsOnBook, PoolFactor, RemainingPoolBalance, MarginalLossRate, CumulativeDefaults_Balance, CumulativeRecoveries, CumulativeNetLossRate and CumulativePrepayments. Dynamic Loss Monthly Data supports monthly monitoring using ReportingDate, MonthlyDefaultRate, MonthlyNetLossRate, CollectionEfficiency, Recoveries_ThisMonth, NewDefaults_Balance, BOP_Balance and EOP_Balance.

Key details:

Item	Description
Main Table	auto_loan_securitisation_data
Vintage Table	static_pool_vintage_data
Monthly Table	dynamic_loss_monthly
DPD Table	dpd_snapshot_history

4. Data Model and Power BI Design

The Power BI model was designed to support both executive reporting and detailed risk analysis. The model includes separate tables for loan-level data, DPD snapshots, vintage performance, dynamic monthly loss metrics, and stress scenario values.

Power Query was used to load and transform the datasets. Relationships were established where required, while some pages used table-specific measures to avoid ambiguity. Measures were stored under relevant home tables such as `auto_loan_securitisation_data`, `static_pool_vintage_data`, `dynamic_loss_monthly` and `Scenario`.

A consistent dashboard style was followed across pages: title at top-left, KPI cards in the top row, main analytical visuals in the middle, and insights or detail tables where useful. This creates a professional report flow similar to a risk management pack.

Key details:

Item	Description
Modelling Approach	Power Query loading and DAX-based calculations
Visual Layout	KPI row + analysis charts + drill-down tables
Design Goal	Clear executive interpretation and analytical depth
Data Grain	Loan-level, state-level, vintage-level and monthly-level

5. Important DAX Measures

Several DAX measures were created to convert raw columns into risk metrics. The most important measures include Total EAD, Total ECL, Average PD, Average LGD, DPD loan counts, baseline and stressed ECL, collection efficiency, net loss rates and vintage performance metrics.

IFRS 9 measures allow the dashboard to evaluate expected losses by stage and risk group. Stress testing measures simulate increased loss exposure under mild and severe conditions. Dynamic monthly measures track operational performance such as recoveries and defaults.

The measures were intentionally kept interpretable so that the business meaning is clear. For example, Mild Stress ECL equals baseline ECL multiplied by 1.10 and Severe Stress ECL equals baseline ECL multiplied by 1.20.

Key details:

Item	Description
Total EAD	SUM(EAD)
Total ECL	SUM(ECL_Provision)
Average PD	AVERAGE(PD_Estimate)
Average LGD	AVERAGE(LGD_Estimate)
Stress ECL	Baseline ECL multiplied by scenario factor

6. Page 1 - Portfolio Overview

The Portfolio Overview page acts as the opening executive dashboard. It presents Total EAD, Total ECL, Total Current Balance and Net Loss as KPI cards. These indicators provide a quick view of the size and risk level of the securitised portfolio.

The page also includes Total EAD by Region, Net Loss by State, IFRS 9 Stage distribution and a loan-level detail table. The regional and state charts help identify exposure and loss concentration. The IFRS 9 stage chart shows whether most loans are performing or credit-impaired.

This page is valuable for senior stakeholders because it answers the first question in any securitisation review: What is the overall portfolio condition? It also provides entry points for deeper pages such as delinquency, ECL and geography.

Key details:

Item	Description
Key KPIs	Total EAD, Total ECL, Current Balance, Net Loss
Main Visuals	Region exposure, state loss, IFRS 9 stage split
Business Use	Executive summary and portfolio concentration view

7. Page 2 - DPD and Delinquency Analysis

The DPD and Delinquency page focuses on early warning signals. It includes KPI cards for 30+ DPD Loans, 60+ DPD Loans, 90+ DPD Loans and Average Delinquency Days. These metrics help identify worsening credit behavior before losses fully materialize.

The DPD Bucket Distribution shows that most loans are in the 0-30 bucket, while smaller groups fall into 31-60, 61-90 and 90+ buckets. The Average Delinquency Days by Region chart highlights regional differences in payment behavior. A loan detail table provides drill-down into individual delinquent accounts.

This page is important because delinquency is a leading indicator of future credit losses. Monitoring DPD buckets helps collection teams prioritize accounts and helps risk teams identify segments requiring closer supervision.

Key details:

Item	Description
30+ DPD Loans	78
60+ DPD Loans	57
90+ DPD Loans	33
Average DPD	21.21 days

8. Page 3 - IFRS 9 ECL Analysis

The IFRS 9 ECL Analysis page evaluates the portfolio using Probability of Default, Loss Given Default and Expected Credit Loss. KPI cards show Total ECL, Total EAD, Average PD and Average LGD.

The donut chart displays ECL distribution by IFRS 9 stage. The scatter plot shows PD versus LGD risk segmentation, with bubble size representing ECL provision and color indicating IFRS 9 stage. This helps identify high-risk loans with high default probability, high loss severity, or large ECL contribution.

The ECL by Region chart shows that some regions contribute more heavily to provisions. The insight from this page is that Stage 3 loans remain the largest contributor to expected credit losses and require focused monitoring.

Key details:

Item	Description
Total ECL	11.66M
Average PD	0.08
Average LGD	0.50
Core Insight	Stage 3 drives majority of ECL

9. Page 4 - Stress Testing and Scenario Analysis

The Stress Testing page models how expected credit losses change under adverse assumptions. Three scenarios were used: Baseline, Mild Stress and Severe Stress. Baseline ECL is the current expected loss level, Mild Stress ECL applies a 10% uplift, and Severe Stress ECL applies a 20% uplift.

The scenario comparison chart shows the increase in expected losses as stress intensity rises. The waterfall chart highlights incremental ECL impact. Stage-wise ECL exposure shows that Stage 3 loans continue to dominate loss provisions under stressed conditions.

This page directly supports the project requirement for stress testing frameworks. It helps management understand the portfolio sensitivity to credit deterioration and supports capital planning, provisioning and investor communication.

Key details:

Item	Description
Baseline ECL	11.66M
Mild Stress ECL	12.83M
Severe Stress ECL	14.00M
Stress Impact	1.17M for mild stress

10. Page 5 - Geographic Risk Analysis

The Geographic Risk Analysis page evaluates state-level and regional risk concentration. KPI cards show Total States, Average PD, Average LGD and Total Regional ECL.

The dashboard includes ECL by State, LGD by State, Regional Risk Ranking and Top 10 High Risk States based on PD. These visuals make it easier to identify states where loss provisions or default probabilities are elevated.

The main business insight is that credit losses are concentrated in a limited number of states and regions. Such concentration risk should be monitored because local economic or collection issues can materially affect portfolio performance.

Key details:

Item	Description
Total States	19
Average PD	0.08
Average LGD	0.50
Total Regional ECL	11.66M

11. Page 6 - Vintage and Static Pool Analysis

The Vintage and Static Pool Analysis page evaluates the seasoning behavior of the portfolio. KPIs include Average Months on Book, Average Pool Factor, Average Marginal Loss Rate, Remaining Pool Balance, Total Recoveries and Average Net Loss Rate.

The Remaining Pool Trend and Pool Factor Decay visuals show the expected runoff of the loan pool over time. Cumulative Default Balance, Recovery Trend, Net Loss Rate Evolution and Prepayment Behavior explain how credit losses, recoveries and amortization develop as loans season.

This page is critical for securitisation because investors often evaluate static pools and vintages to understand whether a pool is performing according to expectations. The observed behavior indicates normal seasoning, with balances declining and losses/recoveries accumulating over time.

Key details:

Item	Description
Avg Months on Book	15.36
Avg Pool Factor	0.68
Avg Marginal Loss Rate	0.03%
Remaining Pool	8.98bn

12. Page 7 - Dynamic Loss and Performance Analysis

The Dynamic Loss and Performance page tracks monthly operational and credit performance. KPI cards include Total Monthly Recoveries, Average Collection Efficiency, Average Monthly Default Rate and Average Monthly Net Loss Rate.

The page includes Monthly Default Rate Trend, Recoveries versus New Defaults, Collection Efficiency Trend and Portfolio Balance Evolution. Together these visuals explain whether collections are stable, defaults are increasing, and balances are declining or spiking across reporting periods.

The page provides a monthly performance lens that complements the static pool and portfolio-level analysis. It is useful for monitoring ongoing servicer performance, recovery effectiveness and monthly loss behavior.

Key details:

Item	Description
Total Monthly Recoveries	4.43M
Avg Collection Efficiency	1.29
Avg Monthly Default Rate	0.04
Avg Monthly Net Loss Rate	0.12%

13. Key Findings

The dashboard reveals that the portfolio is broadly performing but contains concentrated risk in Stage 3 loans, selected states and higher DPD buckets. Stage 3 loans contribute the largest share of expected credit losses.

Delinquency metrics show that most loans are within the 0-30 DPD bucket, while a smaller but important group of accounts fall into 60+ and 90+ buckets. These delinquent loans should be prioritized for collections.

Geographic analysis shows that losses are not evenly distributed. Some regions and states have higher ECL or PD levels. Stress testing confirms that losses increase materially when adverse conditions are applied.

Key details:

Item	Description
Finding 1	Stage 3 loans dominate ECL
Finding 2	90+ DPD accounts require focused attention
Finding 3	Losses are geographically concentrated
Finding 4	Stress scenarios increase provisions

14. Recommendations

The first recommendation is to strengthen monitoring of Stage 2 and Stage 3 loans. These accounts should be reviewed regularly because they have higher expected loss contribution and may materially affect portfolio performance.

The second recommendation is to improve collection strategy for 60+ and 90+ DPD accounts. Early intervention can reduce migration into severe delinquency and lower future ECL.

The third recommendation is to monitor high-risk states and regions through dedicated dashboards, alerts and portfolio limits. Geographic concentration should be reviewed in credit committee meetings. Additional enhancements can include machine learning-based default prediction and automated risk alerts.

Key details:

Item	Description
Recommendation 1	Focus on Stage 3 and high-PD loans
Recommendation 2	Prioritize 60+ and 90+ DPD collections
Recommendation 3	Monitor geographic concentration risk
Recommendation 4	Add predictive analytics in future

15. Conclusion

This project successfully delivers a professional Power BI dashboard for securitisation risk assessment. The solution covers the end-to-end risk lifecycle from portfolio overview to delinquency, IFRS 9 ECL, stress testing, geographic risk, vintage analytics and dynamic monthly performance.

The dashboard demonstrates how data analytics can support decision-making in BFSI and structured finance. It converts raw assumed banking data into actionable insights suitable for risk teams, investors, management and portfolio monitoring functions.

The project also shows practical use of Power BI, DAX, data modelling, financial risk concepts and storytelling. Future development can include predictive default modelling, automated refresh, Power BI service deployment, role-level security and investor reporting packs.

Key details:

Item	Description
Status	Complete
Pages Built	7 dashboard pages
Business Value	Risk monitoring and decision support
Future Scope	Predictive analytics and automated reporting

Appendix - Dashboard Page List and Confidentiality Note

The completed Power BI dashboard contains seven report pages: Portfolio Overview, DPD and Delinquency Analysis, IFRS 9 ECL Analysis, Stress Testing and Scenario Analysis, Geographic Risk Analysis, Vintage and Static Pool Analysis, and Dynamic Loss and Performance Analysis.

All related files, Power BI assets and datasets should be stored in a private repository or private submission location as required by the project instructions. Public sharing of implementation, code, concepts or dashboard files should be avoided unless explicitly permitted by the project owner.

Item	Description
Repository	Private GitHub repository only
Submission	Transfer/share as instructed by project platform
Confidentiality	No public dashboard/code sharing
Deliverable	PBIX file, PDF report and feedback video if required